

Mutation in Nuclear Pore Component NUP155 Leads to Atrial Fibrillation and Early Sudden Cardiac Death

Xianqin Zhang,^{1,2,5} Shenghan Chen,^{1,5} Shin Yoo,^{1,5} Susmita Chakrabarti,^{1,5} Teng Zhang,^{1,5} Tie Ke,^{1,2,5} Carlos Oberti,^{1,3,5} Sandro L. Yong,¹ Fang Fang,^{1,4} Lin Li,¹ Roberto de la Fuente,³ Lejin Wang,^{1,2} Qiuyun Chen,¹ and Qing Kenneth Wang^{1,2,4,*}

¹Department of Molecular Cardiology, Lerner Research Institute, Center for Cardiovascular Genetics, Department of Cardiovascular Medicine, Taussig Cancer Center, Cleveland Clinic, Department of Molecular Medicine, Cleveland Clinic Lerner College of Medicine of Case Western Reserve University, 9500 Euclid Avenue, Cleveland, OH 44195, USA

²Key Laboratory of Molecular Biophysics, College of Life Science and Technology and Center for Human Genome Research, Huazhong University of Science and Technology, Wuhan 430074, China

³Department of Cardiology, Ospedale Italiano Umberto I, Montevideo 11600, Uruguay

⁴Department of Chemistry, Cleveland State University, Cleveland, OH 44107, USA

⁵These authors contributed equally to this work

*Correspondence: wangq2@ccf.org

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SUMMARY

Atrial fibrillation (AF) is the most common form of sustained clinical arrhythmia. We previously mapped an AF locus to chromosome 5p13 in an AF family with sudden death in early childhood. Here we show that the specific AF gene underlying this linkage is *NUP155*, which encodes a member of the nucleoporins, the components of the nuclear pore complex (NPC). We have identified a homozygous mutation, R391H, in *NUP155* that cosegregates with AF, affects nuclear localization of *NUP155*, and reduces nuclear envelope permeability. Homozygous *NUP155*^{-/-} knockout mice die before E8.5, but heterozygous *NUP155*^{+/-} mice show the AF phenotype. The R391H mutation and reduction of *NUP155* are associated with inhibition of both export of *Hsp70* mRNA and nuclear import of *Hsp70* protein. These human and mouse studies indicate that loss of *NUP155* function causes AF by altering mRNA and protein transport and link the NPC to cardiovascular disease.

INTRODUCTION

Internal compartmentalization of eukaryotic cells was a critical stage of evolution which enclosed genetic information inside the nucleus and spatially separated it from the rest of the cell (Hoelz and Blobel, 2004). Transport of macromolecules between the nucleus and cytoplasm is a critical cellular process for eukaryotes which directly affects gene expression and signal transduction (Taddei et al., 2006; Terry et al., 2007). The nuclear pore complex (NPC) is an evolutionarily conserved structure that spans the nuclear envelope (NE) and mediates bidirectional exchange of macromolecules (mRNAs and proteins) between

the nucleus and cytoplasm (Weis, 2003). The NPC was first recognized in the 1960s, but its role as a mediator of nucleocytoplasmic transport was not definitively demonstrated until 20 years later (Stewart et al., 2007). The NPC allows for the passive diffusion of molecules of relative molecular mass of 40,000, but actively transports large macromolecules, such as proteins, RNAs, and ribonucleoprotein particles, through shuttling transport receptors (Gorlich and Kutay, 1999). The NPC is composed of some 30 different nucleoporins that are each present in multiple copies. Extensive electron microscopy and tomography studies in addition to an innovative strategy involving biophysics, proteomics, and sophisticated computation has revealed the general morphology of the NPC (Alber et al., 2007a, 2007b; Beck et al., 2004; Hinshaw et al., 1992). The overall shape and architecture of the NPC is conserved from yeast to higher eukaryotes.

Nucleoporins are the critical molecular components for the assembly and function of the NPC. *NUP155* is a nucleoporin with a calculated molecular mass of about 155 kDa and is a major component of the NPC (Zhang et al., 1999). The first *NUP155* gene was cloned from the rat by Radu et al. in 1993, and was shown to be associated with both nucleoplasmic and cytoplasmic aspects of the NPC (Radu et al., 1993). Nevertheless, the in vivo physiological roles of *NUP155* in mammalian systems are unknown. In this study, we employed both human genetic and mouse gene-targeting approaches to demonstrate the essential roles of *NUP155* in cardiac physiology and further link it to the most common clinical cardiac arrhythmia, atrial fibrillation (AF), and early sudden death.

AF is a cardiac disorder characterized by supraventricular tachyarrhythmia due to uncoordinated atrial activation, a rapid atrial rate of 150–300 beats/min, absence of P waves, presence of rapid oscillations or fibrillatory waves (f waves), and inconsistent R-R intervals on electrocardiograms (ECG) (Fuster et al., 2001). The prevalence of AF is increasing rapidly as the population ages, and reaches more than 8% for people aged 80–89 years (Peters et al., 2002). More than 15% of strokes are caused

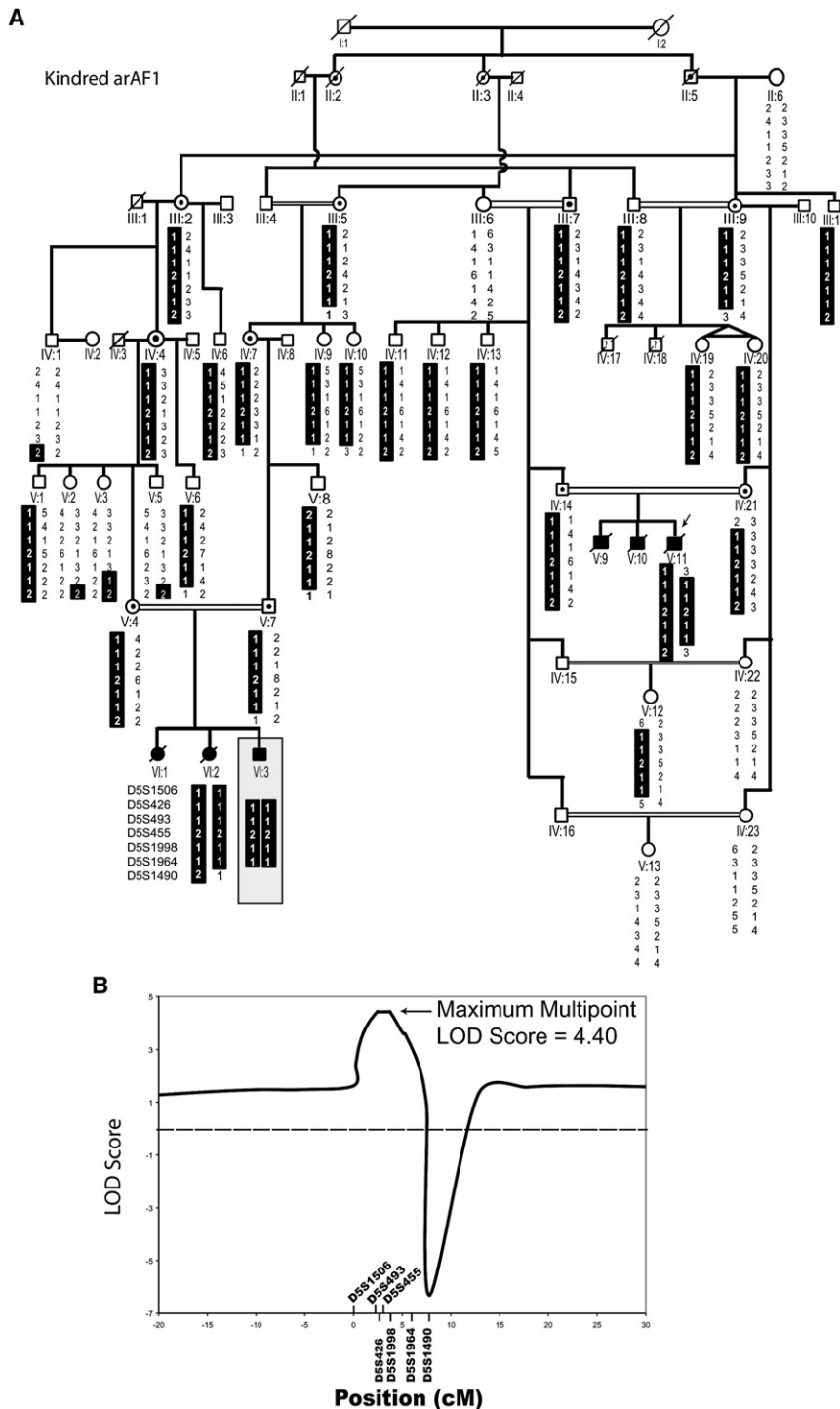


Figure 1. Genotyping and Linkage Analysis of an AF Family, arAF1

(A) Genotyping results are shown for an affected member (boxed VI:3), a newly born male in the arAF1 family, together with the results for other family members reported previously (Oberti et al., 2004). The new affected male patient, VI:3, was genotyped with five linked markers at the arAF1 locus, including D5S493, D5S426, D5S455, D5S1998, and D5S1964. Affected individuals are shown as filled squares (males) and circles (females); normal individuals are indicated by empty symbols; obligate heterozygous carriers are shown with dots in symbols; deceased family members are shown with slashes (/); the proband is indicated by an arrow.

(B) Multipoint LOD score analysis. Genotypes at D5S1506, D5S426, D5S493, D5S455, D5S1998, D5S1964, and D5S1490 were used to determine multipoint LOD scores at chromosome 5p13. D5S1506 was arbitrarily plotted at 0 cM at the abscissa. Other microsatellite markers near the arAF1 locus are indicated along the abscissa from telomere to centromere. Multipoint LOD scores were plotted on the ordinate.

tations in ion channels have been identified in a few families with AF, the genetic basis for the majority of AF cases remains largely unknown (Wang, 2008). We have previously reported a large AF family with an autosomal recessive inheritance pattern and identified the first genetic locus for autosomal recessive AF to chromosome 5p13 (arAF1) (Oberti et al., 2004).

In this study, we provide evidence demonstrating that the AF gene at the 5p13 arAF1 locus is the *NUP155* gene. The results provide a novel non-ion-channel mechanism for the pathogenesis of AF and, to our knowledge, link the NPC to cardiovascular disease for the first time.

RESULTS

Genetic Linkage of Autosomal Recessive AF to Chromosome 5p13 in the Expanded arAF1 Family and Fine Mapping

We previously mapped an autosomal recessive AF locus to chromosome 5p13

in a large arAF1 family (Oberti et al., 2004). After the initial publication, a new affected male was born (individual VI:3; Figure 1A). This patient was genotyped with five markers linked to AF at the arAF1 locus, D5S493, D5S426, D5S455, D5S1998, and D5S1964, and was linked to all five. The maximum two-point logarithm of the odds (LOD) score, at a recombination fraction of 0, increased from the previous 3.05 to 4.04 in the expanded

by AF (Fuster et al., 2001; Peters et al., 2002). AF is also associated with worsening heart failure and increased mortality (Saveleva and John, 2004). Common AF is a complex disease, and its pathogenesis involves multiple genetic factors, environmental factors, and interactions among these factors. Nearly 30% of AF patients present with either an autosomal dominant or recessive familial history (Oberti et al., 2004; Wang, 2008). Although rare mu-

pedigree (see Table S1 available online). The maximum multipoint LOD score increased from 4.10 to 4.40 (Figure 1B). These results strengthen a linkage of AF to chromosome 5p13.

Fine mapping, with single-nucleotide polymorphisms (SNPs) in the region, restricted the 5p13 AF locus between the *NPR3* gene (physical location, 32,747,422) and *PTGER4* (physical location, 40,729,594). A systematic mutational analysis was initiated to identify the AF gene by direct DNA sequence analysis. On the right side of the AF locus, sequence analysis of *OSMR*, *FYB*, and *FLJ39155* (now known as *EGFLAM*; physical location, 38,501,338) identified an SNP in *EGFLAM* that showed a recombination event with AF in the family, further defining the AF locus within a 5.75 Mb interval.

A Missense Mutation in *NUP155* Cosegregates with AF in the arAF1 Family

We systematically analyzed almost all the known and putative genes within the refined arAF1 locus (not including the known disease-causing genes, *SMN1* and *SMN2*, for amyotrophic lateral sclerosis and spinal muscular atrophy, *GDNF* for Hirsch Sprung disease, and *NIPBL* for Cornelia de Lane syndrome). All exons and exon-intron boundaries of the regional candidate genes were sequenced in two DNA samples, one from the affected individual V-11 (Figure 1A) and the other from a normal control individual. The only mutation identified was a homozygous mutation, R391H, in the nucleoporin 155 gene (*NUP155*) (Figure 2A). The R391H mutation occurs in exon 11 at an evolutionarily, highly conserved arginine (R) residue (Figure 2B). In addition, 1700 control individuals did not carry the mutation (data not shown). All affected members of the arAF1 family carried the homozygous R391H mutation, whereas all obligate carriers carried the heterozygous mutation. All normal family members were mutation negative (Figure 3). These results provide strong direct genetic evidence that the R391H is a pathogenic mutation that causes AF in the family.

Mutation R391H Affects Nuclear Envelope Localization of *NUP155* Protein and Nuclear Envelope Permeability

We hypothesized that incorrect NE localization of *NUP155*, mediated by the R391H mutation, is a mechanism for the pathogenesis of AF. To test the hypothesis, we evaluated the nuclear localization of wild-type (WT) and R391H mutant *NUP155* proteins in COS7 cells. COS7 cells were transiently transfected with mammalian expression constructs for enhanced green fluorescent protein (EGFP)-tagged WT or R391H mutant *NUP155* proteins. Wild-type *NUP155* protein localized, in a strong, tight, and punctuated manner with many NE foci (Figure 4A). In contrast, R391H mutant *NUP155* protein failed to visibly accumulate at various NE foci and was diffusely distributed toward and in the cytoplasm. Identical results were obtained by immunostaining studies of COS7 cells transfected with either Flag-tagged WT or R391H mutant *NUP155* constructs (Figure 4B). These results suggest that the R391H mutation in *NUP155* affects the targeting of *NUP155* protein to specific sites (presumably the NPC) on the NE. The functional effect of the R391H mutation was tested by a newly developed assay for NE permeability (Grote and Ferrando-May, 2006). Compared to HeLa cells without transfection or cells with transient expres-

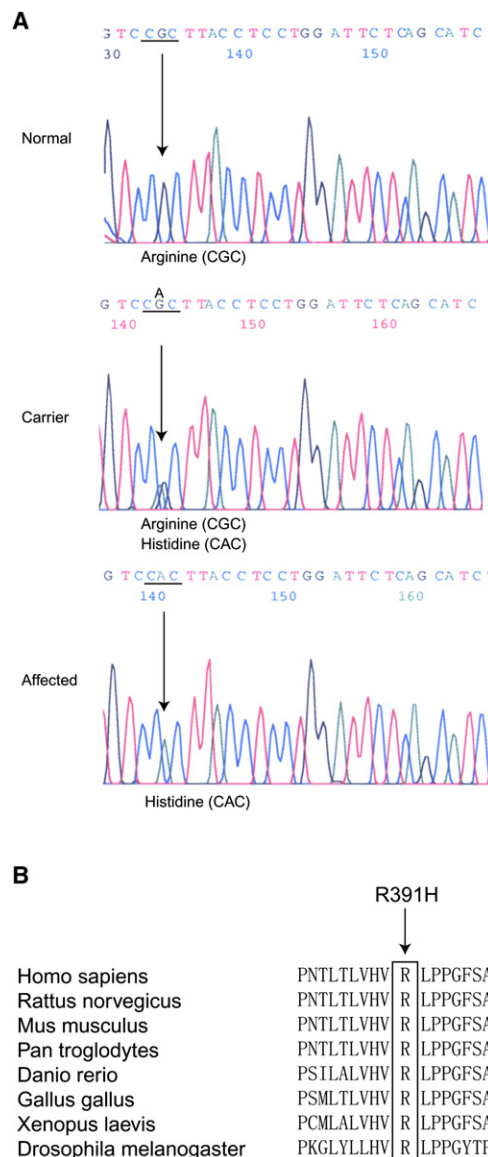


Figure 2. Identification of a Novel R391H Mutation in *NUP155* Associated with AF

(A) DNA sequences of exon 11 of *NUP155* from a normal individual, an obligate carrier, and an affected family member.

(B) The R391H mutation is located in exon 11 and occurs at an arginine (R) residue that is highly evolutionarily conserved.

sion of WT *NUP155*, cells with R391H mutant *NUP155* showed significantly reduced NE permeability (Figures 4C and 4D). The R391H mutation may act by a loss-of-function mechanism consistent with the common mechanism of most recessive human diseases.

Heterozygous *NUP155*^{+/-} Knockout Mice Are Associated with AF

To further strengthen the link between *NUP155* and AF, we developed a knockout mouse line targeting *NUP155* by the

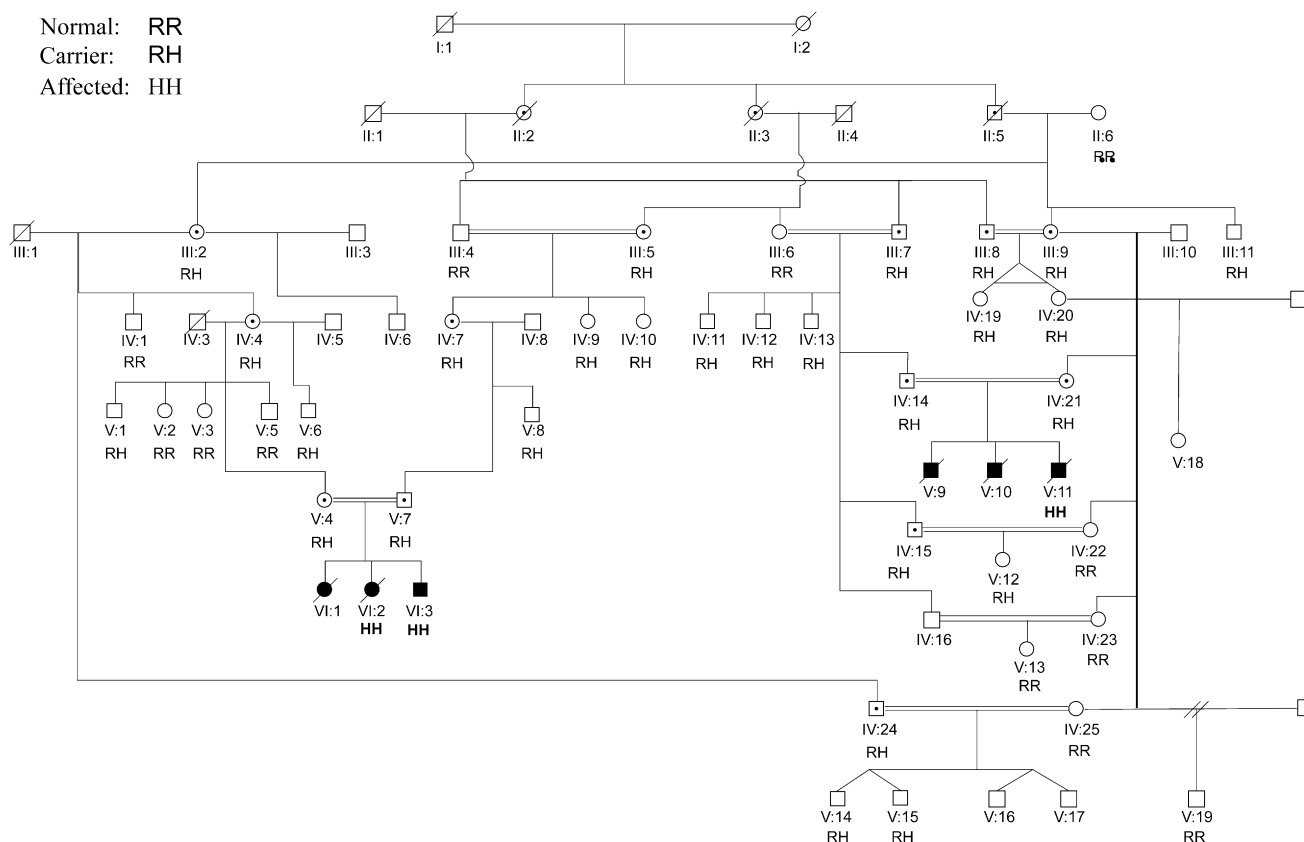


Figure 3. Cosegregation of *NUP155* Mutation R391H with AF in the Family
R, wild-type R391 allele; H, mutant H391 allele.

gene-trapping strategy. The trapping vector was inserted in intron 5 of the mouse *NUP155* gene, which results in a truncated mNUP155 protein of only 217 amino acids (the normal NUP155 contains 1391 amino acids) (Figure 5A). Homozygous *NUP155*^{-/-} mice died during embryogenesis before embryonic day 8.5, whereas heterozygous *NUP155*^{+/-} mice were viable. Semiquantitative RT-PCR and western blot analyses showed that the expression levels of mouse *NUP155* mRNA and protein were significantly reduced by about 50% in heterozygous *NUP155*^{+/-} knockout mice compared with wild-type littermate control mice (Figures 5B and 5C). This result was confirmed using immunostaining with an anti-NUP155 antibody (Figure 5D). The immunostaining study also demonstrated that NUP155 was strongly expressed in atrial myocytes (Figure 5D). Continuous telemetry ECG recordings from conscious, freely moving mice revealed significant similarities in cardiac phenotype between human AF patients and *NUP155*^{+/-} mice (Figure 6). The observed AF phenotype can sustain for 10 min or longer. The main AF-related phenotypes include (1) absence of discrete P waves, and (2) irregular R-R intervals (Figure 6). These results indicate that downregulation of *NUP155* expression by haploinsufficiency (50%) causes AF in mice.

Mutant *NUP155* Reduces Atrial Action Potential Duration and Inhibits Export of *Hsp70* mRNA and Nuclear Import of *Hsp70* Protein

Histological analysis of *NUP155*^{+/-} knockout mice did not reveal any overt structural abnormalities in heart and skeletal muscle (data not shown), the two organs demonstrating the highest expression levels of *NUP155* (Zhang et al., 1999). Immunostaining with cardiac-specific α -actinin of *NUP155*^{+/-} hearts did not reveal obvious differences in the morphology of cardiomyocytes and the atrial tissue architecture (Figure 7A). We then examined *NUP155*^{+/-} atrial myocytes for electrophysiological alternations. Compared to *NUP155*^{+/+} cells, atrial myocytes from *NUP155*^{+/-} mice showed significantly shortened action potential duration (Figure 7B), which is involved in induction and maintenance of AF (Nattel, 1999; Van Wagoner and Nerbonne, 2000).

To identify a molecular mechanism by which mutant NUP155 causes AF, we tested the hypothesis that NUP155 mutation R391H or the reduction of NUP155 may affect mRNA export of *Hsp70*. Export of *Hsp70* mRNA is inhibited by siRNA knockdown of expression of hCG1, which forms a complex with NUP155 and hGle1 (Kendirgi et al., 2005). RT-PCR showed that HeLa cells with transient expression of wild-type NUP155 increased export of *Hsp70* mRNA, but cells with mutant NUP155 exhibited

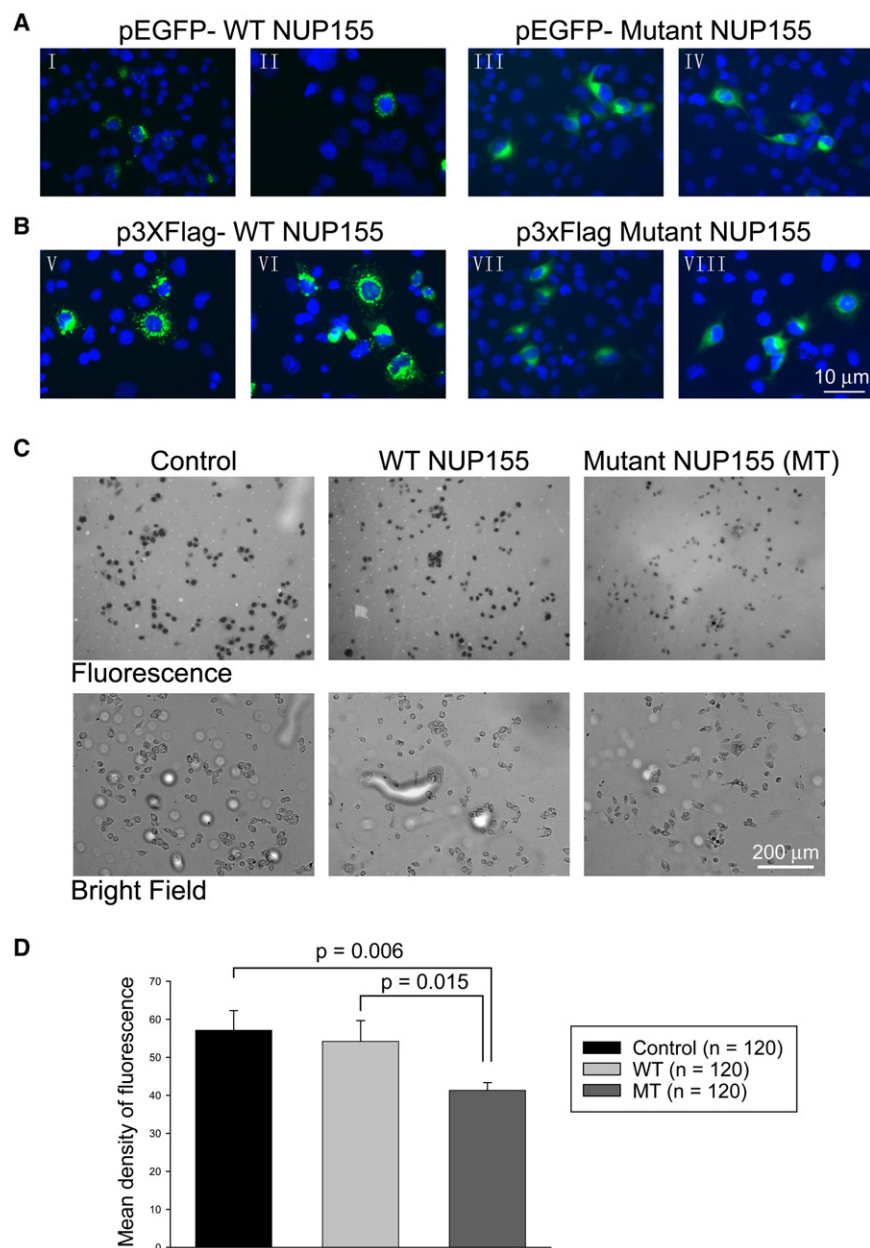


Figure 4. Mutation R391H Affects the Localization of NUP155 on the Nuclear Envelope and Reduces the Permeability of the Nuclear Envelope

(A) EGFP-NUP155.

(B) Flag-NUP155. Wild-type NUP155 (WT) is concentrated in foci at the nuclear envelope, whereas R391H mutant NUP155 failed to visibly accumulate in foci at the nuclear envelope and was diffusely distributed throughout the cytoplasm.

(C) Nuclear envelope permeability assays by influx of Texas red-labeled dextran (70 kDa) into nuclei of semipermeabilized HeLa cells. Top panel, fluorescence images; bottom panel, images of bright field from the same slides.

(D) Quantification of fluorescence intensity from images from permeability assays. A total of 120 fluorescence images (40×) was analyzed for each sample. Mean ± SEM: 57.09 ± 5.21 for control cells without transfection, 54.18 ± 5.46 for cells with transient expression of WT NUP155, and 41.30 ± 2.06 for cells with transient expression of mutant NUP155. The results were analyzed by one-way ANOVA analysis.

but no significant abnormalities in either mRNA export or protein nuclear import were detected (data not shown).

DISCUSSION

Our studies in both humans and mice provide strong evidence for the role of a nucleoporin, NUP155, in the physiology of the heart and the pathogenesis of AF, the most common clinical cardiac arrhythmia. Multiple lines of evidence indicate that *NUP155* is the AF gene on chromosome 5p13. First, we identified significant genetic linkage between markers flanking *NUP155* and AF in a large family with single and multipoint LOD scores of 4.04 and 4.40, respectively. Second, we identified a novel homozygous mutation, R391H in *NUP155*,

that cosegregates with AF in the AF family, but does not exist in 1700 controls. Third, we demonstrated that the R391H mutation affects the NE localization of NUP155 and reduces NE permeability. Fourth, we demonstrated strong expression of NUP155 protein in atrial myocytes. Finally, heterozygous *NUP155*^{+/-} knockout mice showed the AF phenotype, which supports the conclusion that *NUP155* is a disease-causing gene for AF.

Additional evidence supports a role of *NUP155* in the pathogenesis of AF. The human *NUP155* gene was reported to display a tissue-specific expression pattern (Zhang et al., 1999). Northern blot analysis identified a 5.4 kb transcript that showed the highest expression level in two organs, the heart and skeletal muscle (Zhang et al., 1999). Immunostaining studies also

significant reduction of mRNA export (Figure 7C). Similarly, cardiomyocytes from *NUP155*^{+/-} mice showed significantly reduced export of *Hsp70* mRNA compared to those from *NUP155*^{+/+} mice (Figure 7D). These results demonstrate that NUP155 dysfunction significantly inhibits export of *Hsp70* mRNA to the cytoplasm. Immunostaining with an anti-Hsp70 antibody showed that overall expression levels of Hsp70 protein were reduced in HeLa cells with transient expression of mutant NUP155 protein (Figure 7C) or in *NUP155*^{+/-} myocytes (Figure 7D), consistent with the results of reduced export of *Hsp70* mRNA from RT-PCR analysis. Interestingly, nuclear import of Hsp70 protein was also inhibited in HeLa cells with transient expression of mutant NUP155 protein (Figure 7C) or in *NUP155*^{+/-} myocytes (Figure 7D). Similar studies were carried out for Hsp27,

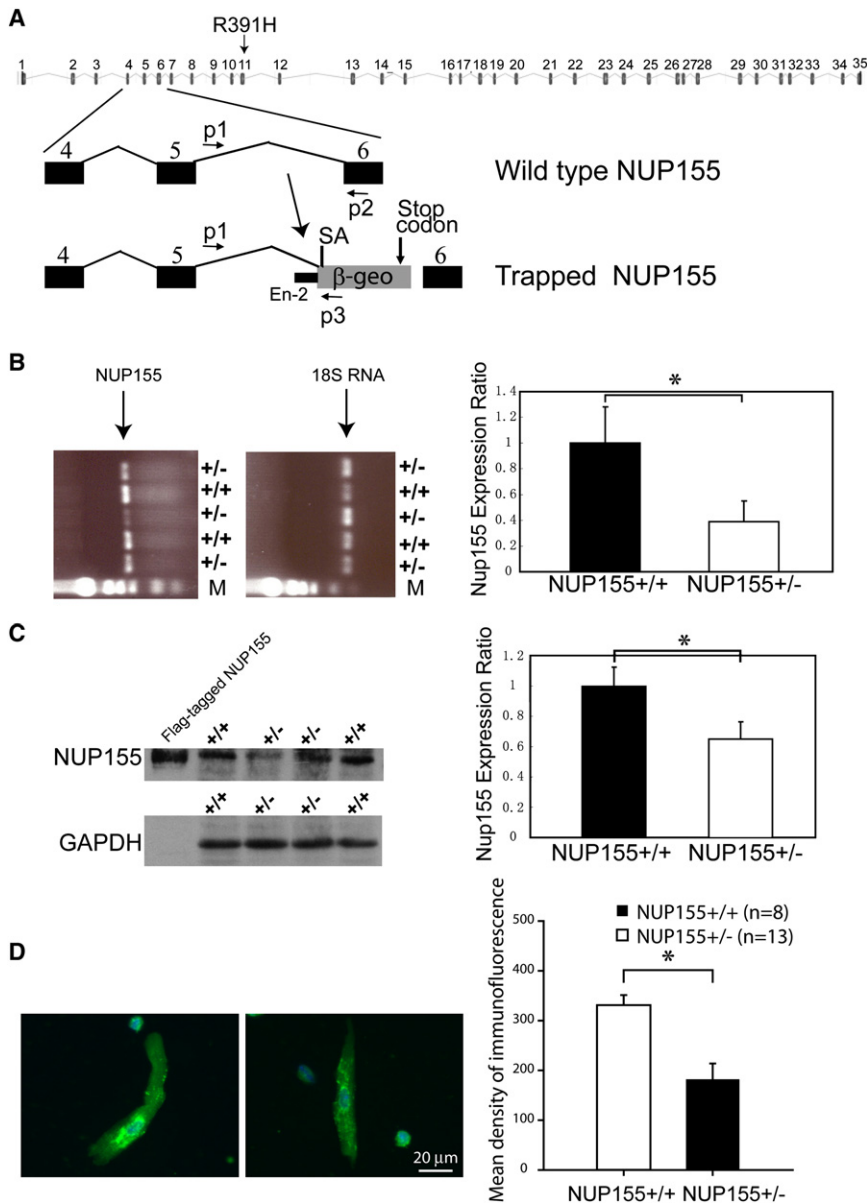


Figure 5. Characterization of Heterozygous *NUP155*^{+/-} Knockout Mice

(A) Strategy for developing a gene-trapping mouse line targeting *NUP155*. The β -geo insertion vector is in intron 5. PCR with primer P1 (5'-ATG GAT GGA GTT AGA TCC CTT AGA C-3') and P2 (5'-AAG ATT CTG CCA TTA TCC GTG GAC-3') identifies the wild-type *NUP155* allele, whereas the mutant, trapped *NUP155* allele is detected by PCR with primers P1 and P3 (5'-GTT TCC TTG AAG GAC TCC AAT-3'). SA, splicing acceptor. (B) Semiquantitative RT-PCR analysis showed reduced expression of *NUP155* by 2-fold in heterozygous *NUP155*^{+/-} mice compared to wild-type *NUP155*^{+/+} mice.

(C) Western blot analysis with an anti-NUP155 antibody demonstrated 50% reduction of NUP155 expression in heterozygous *NUP155*^{+/-} mice compared to wild-type *NUP155*^{+/+} mice.

(D) Immunostaining with an anti-NUP155 antibody showed expression of NUP155 in isolated atrial myocytes. Expression of the NUP155 protein was reduced by 50% in heterozygous *NUP155*^{+/-} mice compared to wild-type *NUP155*^{+/+} mice. The data are shown as mean $\pm$ SEM.

demonstrated strong expression of NUP155 in isolated atrial myocytes (Figure 5). These results are consistent with our conclusion that defects in *NUP155* are associated with AF and provide an explanation as to why the heart is the major organ affected by the loss of function of an essential nucleoporin gene, such as *NUP155*.

To date, our understanding of the genetic basis of AF is mostly limited to genes encoding ion channels. Rare mutations associated with AF have been reported in genes encoding potassium channels, and somatic mutations in the *GJA5* gene encoding connexin40 were identified in tissues from three sporadic AF patients (Wang, 2008). However, the genetic basis for the majority of AF cases remains largely unknown. Positional cloning, based on genome-wide linkage analysis, in large families is the most effective and convincing approach for identifying disease-caus-

ing genes. The initial AF-causing mutation in *KCNQ1* was identified by a positional candidate gene approach, whereas other ion-channel genes associated with AF were identified by the candidate gene approach in sporadic patients or small nuclear families (Wang, 2008). *NUP155* is another AF gene identified by a positional cloning approach. Most importantly, results from this study provide a novel non-ion-channel mechanism underlying the pathogenesis of AF. After the submission of this manuscript, a heterozygous mutation in the *NPPA* gene encoding atrial natriuretic peptide was identified in a family with autosomal dominant AF (Hodgson-Zingman et al., 2008), which provides further support for a non-ion-channel basis of AF.

Our data predict a likely cellular mechanism for the pathogenesis of AF. NUP155 has been demonstrated to interact with an mRNA export factor, Gle1 (Kendirgi et al., 2005; Rayala et al., 2004). The N-terminal 29 amino acid residues of hGle1 are responsible for the interaction with NUP155 and for the targeting of hGle1 to the NPC (Rayala et al., 2004). Further studies showed that NUP155, hGle1, and nucleoporin hCG1 form a heterotrimeric complex, and siRNA knockdown of hCG1 expression is associated with inhibition of export of *Hsp70* mRNA and nuclear accumulation of *Hsp70* mRNA (Kendirgi et al., 2005). Based on these results, we predict that *NUP155* deficiency may cause molecular remodeling of gene expression by affecting mRNA export of key atrial genes from the nucleus

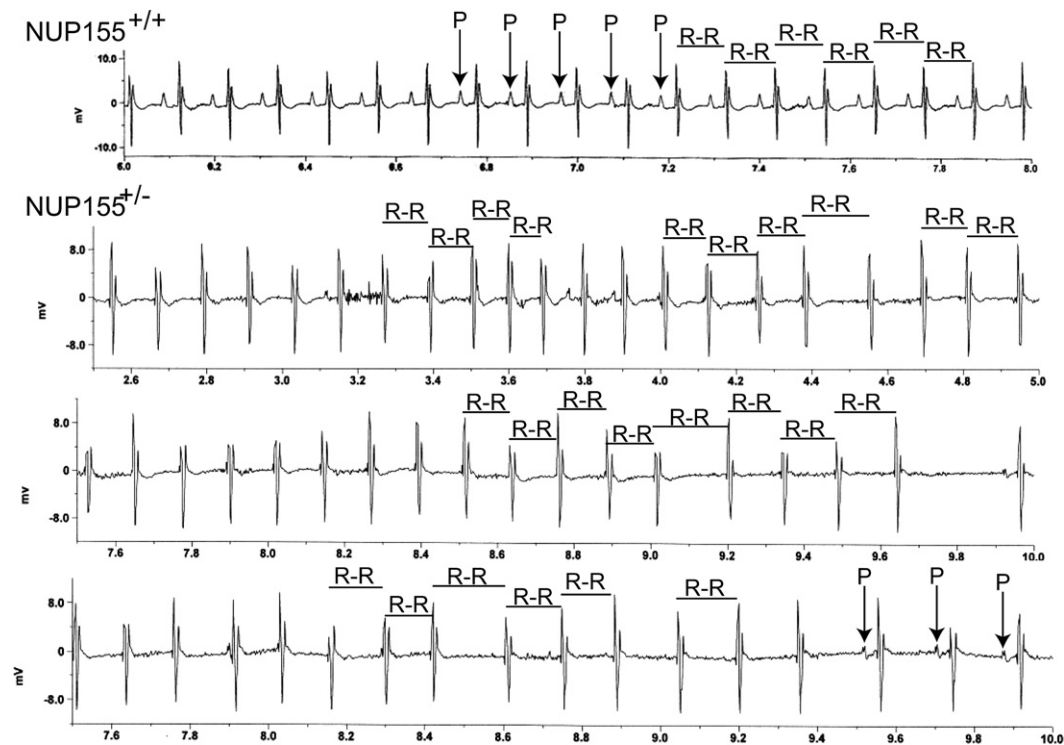


Figure 6. Detection of the Phenotype of AF in Heterozygous *NUP155*^{+/-} Knockout Mice

Telemetric ECG recordings detected the development of atrial fibrillation in *NUP155*^{+/-} mice, but not in *NUP155*^{+/+} mice. Traces are shown as time frame durations in seconds with ECG signals in mV. One representative trace from *NUP155*^{+/+} mice and three traces from three different *NUP155*^{+/-} mice are shown.

to the cytoplasm. Thus, *NUP155* may act as a key upstream gene that causes AF by regulating the expression of downstream genes by controlling mRNA nucleocytoplasmic export. The specific downstream target genes of *NUP155*, in the atria, remain to be identified. However, *Hsp70* may be an interesting candidate gene. Indeed, the R391H *NUP155* mutation and the reduction of *NUP155* in heterozygous mice inhibit export of *Hsp70* mRNA and nuclear import of *Hsp70* protein (Figure 7). Multiple reports link *Hsp70* to AF. Patients with high *Hsp70* expression levels have a lower incidence of postoperative AF (Kampinga et al., 2007; Mandal et al., 2005; St. Rammos et al., 2002). An M439T substitution in *Hsp70* was associated with an increased risk of postoperative AF (Afzal et al., 2008). Cardiomyocytes from *Hsp70* knockout mice exhibit delayed calcium transient decline and reduced sarcoplasmic reticulum calcium content (Kim et al., 2006). An abnormality in intracellular Ca^{2+} homeostasis may be associated with AF (Chou et al., 2008). Heat shock proteins such as *Hsp70* play important roles in protein folding, trafficking, and cell signaling. We propose that *NUP155* may indirectly modulate maturation and trafficking of calcium handling proteins and ion channels through regulation of mRNA and protein transport of *Hsp70* or other important atrial proteins, which may provide an explanation for our finding that the action potential duration from *NUP155*^{+/-} atrial myocytes is shorter than that from control *NUP155*^{+/+} cells (Figure 7B).

The function of *NUP155* has been explored with a variety of biochemical and cell biological approaches. The human and mouse *NUP155* genes were cloned by Zhang et al. in 1999 and 2002, respectively (Zhang et al., 1999, 2002). *NUP155* is evolutionarily conserved from yeast, *Arabidopsis*, *Caenorhabditis elegans*, *Drosophila*, mice, and rats to humans. Immunodepletion and siRNA studies, in *C. elegans* embryos and *Xenopus laevis* egg extracts, indicated that *NUP155* has an essential function in the formation of the NE (Franz et al., 2005). The results in this study indicate that *NUP155* is involved in both export of *Hsp70* mRNA and nuclear import of *Hsp70* protein. However, the in vivo physiological roles of *NUP155* in mammalian systems are unknown. Here we generated a knockout mouse line targeted to *NUP155*. Homozygous *NUP155*^{-/-} mice died before E8.5, suggesting that *NUP155* is essential for embryogenesis. Heterozygous *NUP155*^{+/-} mice showed an interesting phenotype of AF, the most common clinical cardiac arrhythmia. The mouse data were further extrapolated with genetic studies of human patients showing association of *NUP155* with AF. These results reveal a novel role for *NUP155* in cardiac physiology and link the NPC to cardiovascular disease. *NUP155* deficiency may affect the formation of the NE, and delay or block mitosis, which may reduce myocyte survival. In the long run, myocyte apoptosis and cardiac fibrosis may occur, creating substrate for the initiation and maintenance of AF (Ehrlich et al., 2008).

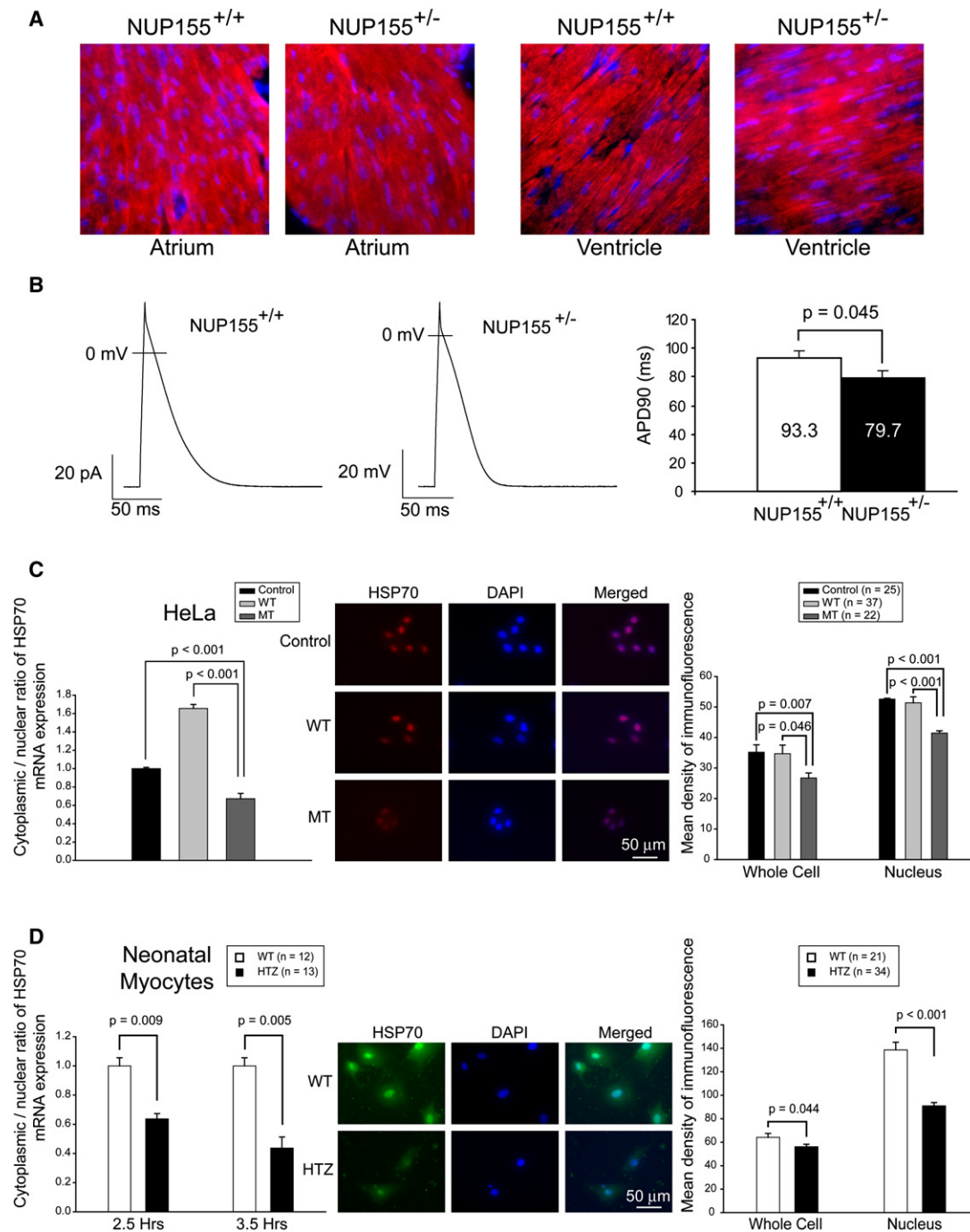


Figure 7. NUP155 Defects Reduce Atrial Action Potential Duration, Export of *Hsp70* mRNA, and Nuclear Import of Hsp70 Protein

(A) Immunostaining of atria and ventricles from *NUP155*^{+/+} and *NUP155*^{+/-} mice for α -actinin.

(B) Representative action potential recordings from *NUP155*^{+/+} and *NUP155*^{+/-} atrial cells. Action potential duration was measured at 90% repolarization (APD₉₀). The mean averages from each cell (n = 12 cells; 3 mice total for each group) were then pooled together for a group average and summarized.

(C) Analyses of export of *Hsp70* mRNA and import of Hsp70 protein in HeLa cells transiently transfected using wild-type (WT) or mutant (MT) *NUP155* expression plasmids. Cells without transfection were used as a control. Forty-eight hours after transfection, cells were heat shocked for 2.5 hr, and total RNA from nuclear and cytoplasmic fractions was isolated and reverse transcribed with an oligo(dT)₁₅ primer. Quantitative real time RT-PCR analysis was then used to estimate the levels of *Hsp70* mRNA in the cytoplasm and nucleus (left panel). All the samples were analyzed in triplicate and the results were replicated in three independent experiments. The heat-shocked cells were stained with an anti-Hsp70 antibody (red) and DAPI for nuclei (blue) (middle panel). Fluorescence intensity in the whole cell or in the nucleus alone was quantified and is shown as a graph (right panel). n, number of images analyzed.

Clinical findings associated with the *NUP155* mutation is of particular interest, as *NUP155* mutation carriers were associated with a presentation of in utero and neonatal AF as well as early sudden death (Oberti et al., 2004). The proband (V:11) and AF patients V:9, V:10, and VI:1 (Figure 1A) all died suddenly at the age of 15, 3, 3, and 18 months, respectively. The sole survivor, at the time of the prior publication (Oberti et al., 2004), AF patient VI:2 (Figure 1A), died recently at the age of 20 months. The early deaths in the family may be related to AF-induced ventricular tachyarrhythmias and nonischemic cardiomyopathies (Knight, 2003; Shinbane et al., 1997).

In humans, only homozygous carriers with mutation R391H were affected with AF, whereas heterozygous carriers were asymptomatic. On the other hand, heterozygous *NUP155*^{+/-} mice developed AF. This may be due to a possibility that the R391H mutant *NUP155* allele is a weaker mutation than haploinsufficiency. The R391H mutant *NUP155* protein may retain some residual nucleoporin activity, and causes AF in the human, only when it is in the homozygous state. Future studies are needed to clarify this issue.

Laramie et al. identified a putative locus that influences resting heart rate, with suggestive evidence of linkage to markers on chromosome 5p13 (LOD score of 1.96) (Laramie et al., 2006) which overlaps with our 5p13 AF locus. It is also interesting to note that heterozygous *NUP155*^{+/-} mice show a variable heart rate (Figure 6). It is possible that *NUP155* variants will also be associated with variation of the human resting heart rate.

In conclusion, our findings identify a novel molecular, non-ion-channel determinant for AF, and extend the understanding of the nuclear pore complex to cardiovascular physiology. To our best knowledge, this study is the first to link a nucleoporin defect to cardiovascular disease. The molecular identification of *NUP155* as an AF gene may facilitate the development of new diagnostic and therapeutic tools for prevention and treatment of AF.

EXPERIMENTAL PROCEDURES

Study Subjects and Linkage Analysis

Study subjects involved were enrolled as described previously (Oberti et al., 2004). The new patient VI:3 (Figure 1A) was born on February 14, 2007 with AF and atrial tachycardia. The patient is currently in sinus rhythm and on propafenone, propranolol, and aspirin. This study was approved by the Cleveland Clinic Institutional Review Boards on Human Subject Research and informed written consent was obtained for each study subject. Whole blood (10–20 ml) was obtained from each subject and human genomic DNA was isolated from the whole blood using the DNA isolation kit for mammalian blood (Roche Diagnostic).

Genotyping of polymorphic microsatellite markers was carried out as described previously (Wang et al., 1995, 1996). Phenotyping data, genotyping data, and pedigree information were used for pairwise linkage analysis using the Fastlink software package (Cottingham et al., 1993). Single-point and multipoint linkage analyses were performed assuming an autosomal recessive pattern of inheritance, a disease-allele frequency of 0.001, and penetrance of 0 for carriers and noncarriers and 0.99 for homozygous affected individuals.

Logarithm of the odds (LOD) scores were calculated as described previously (Oberti et al., 2004).

Mutational Analysis

Mutation screening was carried out using direct DNA sequence analysis. All exons and exon-intron boundaries of each candidate gene were PCR amplified. The PCR products were purified using the PCR purification kit (QIAGEN) and subjected to DNA sequencing analysis using both forward and reverse primers using the ABI BigDye Terminator Cycle Sequencing kit v1.1 with an ABI PRISM 3100 Genetic Analyzer (Applied Biosystems).

Cell Culture and Immunostaining

COS7 cells were maintained in Dulbecco's minimum essential media (DMEM) supplemented with 10% heat-inactivated fetal bovine serum (Invitrogen). Transfection of COS7 cells (10⁶ cells) was carried out with 2 µg of each mammalian expression construct for *NUP155*, including p3xFlag-WT-*NUP155* and p3xFlag mutant-*NUP155* using Lipofectamine 2000 (Invitrogen). Cells were plated on coverslips in a 10 cm plate for 12 hr before the transfection. Forty-eight hours after the transfection, immunostaining was carried out with the M2 anti-FLAG monoclonal antibody (Sigma) as the primary antibody and a FITC-conjugated anti-mouse antibody (Sigma) as the secondary antibody. Cells in coverslips were fixed in 4% paraformaldehyde/PBS for 15 min at room temperature, treated with 0.5% Triton X-100, and then blocked in 5% milk in PBS + Tween 20 (PBST) for 1 hr. The coverslips were then incubated with the M2 anti-FLAG monoclonal antibody (1:1000 dilution in 5% milk in PBST) for 12 hr at 4°C, washed with PBST, and incubated with the secondary FITC-conjugated anti-mouse antibody (1:1000 dilution in 5% milk in PBST) for 1 hr at room temperature. Then, 1–2 drops of Vectashield containing 4',6-diamidino-2-phenylindole (DAPI) was added and the coverslips were placed on the slides, sealed with nail oil, air-dried, and analyzed under an IX81 motorized inverted research fluorescence microscope (Olympus).

COS7 cells were also transfected with expression constructs for EGFP-tagged *NUP155*, pEGFP-WT-*NUP155*, or pEGFP mutant-*NUP155* as described above. The cells on the coverslips were examined directly under an IX81 motorized inverted research fluorescence microscope (Olympus).

Quantitative RT-PCR Analysis

Total RNA was prepared from heart tissue of the heterozygous *NUP155* knockout and wild-type mice. Total RNA was then isolated using the TRIzol reagent (Invitrogen). Double-stranded complementary DNA (ds-cDNA) was synthesized from 16 µg of total RNA using Stratagene's RT-PCR kit. For semiquantitative RT-PCR, we designed the primers to amplify exons 23 and 24 of the mouse *NUP155* gene. The forward primer is 5'-GAA TTT CAA GAG CAG CTG AAG ATC A-3' and the reverse primer is 5'-CTT GCC TAT ACT GAG CAC AGA CAC-3'. 18S rRNA primers were used as an internal control (Tian et al., 2004a). The number of cycles for PCR was optimized for each sample to obtain data in the linear range for quantification. The PCR profile was 25 cycles (for *NUP155*) or 18 cycles (for the 18S RNA) of denaturation (94°C for 30 s), annealing (55°C for 30 s), and extension (72°C for 1 min) and a final extension step of 10 min at 72°C. The PCR products were separated on a 1% agarose gel.

Western Blot Analysis

Western blot analysis was carried out with an anti-*NUP155* antibody raised against C-terminal residues (VQAITGNFKSLQAKLERLH; a kind gift from S.R. Wente; Kendirgi et al., 2005).

Generation of *NUP155* Knockout Mice

To demonstrate that *NUP155* is a causative gene for AF, we developed a knockout mouse line targeting *NUP155*. A gene-trapping embryonic stem (ES) cell line for *NUP155*, RRD107, was identified from the BayGenomics

(D) Analyses of export of *Hsp70* mRNA and import of *Hsp70* protein in neonatal myocytes isolated from wild-type *NUP155*^{+/+} (WT) and heterozygous *NUP155*^{+/-} (HTZ) mice. Neonatal myocytes were heat shocked for 2.5 and 3.5 hr for RT-PCR analysis or 2.5 hr for immunostaining and analyzed as described in (C). Green signal, *Hsp70* protein; blue signal, DAPI staining for nuclei; n, number of mice used for the assay. The data are shown as mean ± SEM.

database (<http://baygenomics.ucsf.edu/>). The ES cell line was recovered and resequenced to confirm its identity. The Bay Genomics/University of California, Davis facility performed blastocyst injection of the ES cells, which yielded 13 chimeras (11 males and 2 females) and successful germline transmission.

Isolation of Mouse Atrial Myocytes, Measurements of Action Potential Duration, and Immunocytochemistry

The procedures for isolation of atrial myocytes from mice were modified from Rose et al. (2004), and detailed protocols are included in Supplemental Data. The action potential duration of isolated atrial myocytes was measured as described previously (Tian et al., 2004b; Yong et al., 2007).

Isolated atrial myocytes were seeded onto carbon-coated glass coverslips and washed with PBS (pH 7.4) (Sigma). The myocytes were fixed in 10% neutral buffered formalin (Sigma) for 15 min, washed three times with PBS, and permeabilized by incubation in PBS containing 0.1 Triton X-100 for 10 min. After washing three times with PBS, the myocytes were blocked in 10% normal goat serum (NGS; Sigma) in PBS for 60 min, and incubated with a 1:100 concentration of the anti-NUP155 antibody diluted with PBS containing 1% BSA and 10% NDS in a humid box at -4°C overnight. After washing three times with PBS, the myocytes were incubated with goat anti-rabbit IgG conjugated to FITC (Sigma) in a humid box at room temperature in the dark for 1 hr. After washing three times with PBS, the myocytes were mounted with Vectashield mounting media (Vector Labs), sealed with nail polish, and visualized using an epifluorescent microscope (BX61; Olympus). The mean intensity of the protein signal in a myocyte was measured using Image Pro 6.1 (Media Cybernetics). Adobe Photoshop CS and Illustrator CS (Adobe Systems) were used for processing images.

Mouse ECG Recordings

ECG recordings in mice were carried out as described previously (Tian et al., 2004b; Zhang et al., 2007). Mice (5–10 months, 30–35 g) were anesthetized with 0.015 ml/g body weight Avertin (2.5%; Sigma-Aldrich). TA10ETA-F20 transmitters (Data Sciences International) were surgically implanted into the peritoneal cavity of the *NUP155^{+/-}* mice. Nontransgenic littermates (*NUP155^{+/+}*) were used as controls. After the surgery, the mice were allowed to recover for 1 week prior to recording of ECG. The Data Sciences telemetry system was used to continuously monitor and record ECG of conscious, unrestrained mice using Data Sciences data acquisition ECG software and saved on a hard disk for later analysis. Signals below 248 MHz were filtered out.

Nuclear Envelope Permeability Assays

The permeability of the nuclear envelope was measured using a newly developed assay as described (Grote and Ferrando-May, 2006). In brief, cells on coverslips (15,000 cells/coverslip) were permeabilized using a low concentration of digitonin (6 $\mu\text{g}/\text{ml}$). These semipermeabilized cells retained the intact nuclear membrane. The cells were incubated with cytosolic S-20 extract and a fluorescent reporter, 70 kDa Texas red-labeled dextran (Molecular Probes). Cells were then visualized under a microscope. Images were taken and fluorescence intensity in the nuclei was quantified using Image Pro 6.1 (Media Cybernetics).

Assays for Export of *Hsp70* mRNA and Nuclear Import of *Hsp70* Protein

Export of *Hsp70* mRNA was assayed using RT-PCR analysis of cytoplasmic and nuclear RNA as described (Kendirci et al., 2005). The RT-PCR data were analyzed as described (Pfaffl, 2001). Details are in Supplemental Data.

Nuclear import of *Hsp70* protein was measured using immunostaining analyses as previously described (Zeng et al., 2004). A mouse monoclonal antibody (Assay Designs, no. SPA-810) was used for a nuclear import assay in HeLa cells, and a rabbit polyclonal antibody (Assay Designs, no. SPA-812) was used for the assay in mouse neonatal myocytes.

Statistical Analysis

Data are shown as mean values $\pm$ SEM. Statistical analysis was performed using ANOVA and two-tailed Student's *t* tests to compare means, and significance was set at the *p* value of < 0.05 .

SUPPLEMENTAL DATA

Supplemental Data include one table and Supplemental Experimental Procedures and can be found with this article online at [http://www.cell.com/supplemental/S0092-8674\(08\)01314-7](http://www.cell.com/supplemental/S0092-8674(08)01314-7).

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